Brain MRI Tumor Segmentation and Quantitative Analysis Using Classical Image Processing

Mid Semester Evaluation Report

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Repo Link:-
https://github.com/Rushin-Mod/DIP_2026_P20_G9

I. Motivation behind this Project:-

Brain Tumors are one of the most deadliest forms of ailments a person can go through and an early screening and detection can help a person be safe from danger by starting any necessary therapies, malignant tumors could have adverse effects too, messing up a person's ability to think and remember. With Time, there has also been a rise in cases related to brain tumors, so we wanted to implement something which could behave like a detection device for anyone who might need it.

We also wanted to make sure that nobody would have to face the likelihood of getting a false positive result, which would cause panic within everyone associated with the person & We would also like to make sure that nobody gets a false negative result so they might not get a checkup after for a while and then it gets too late and the damage is irreversible, which is a case when the tumor actually becomes too large and causes head trauma. We want to help the society in a form that would be massively helpful for a lot of the populace. We aim to do good towards our fellow people through this idea, so that people do not have to see their friends and family in a hurtful state.

II. Problem Definition:-

If we were to define the Problem statement in 1-2 lines, it would be as follows:

> *"Using Classical Image Processing Methods like Filtering and Segmentation to detect Brain Tumors in MRI Scans."*

III. Work done throughout the Weeks:-

For Week 1, We basically went through the different topics selected this one and looked into the Literature associated with it, Most Recent Papers are based on Topics of CV and ML over Classical Methods.

We had to look further back in time so that we could find more papers in relation to the kind of topic we were going for, We tried to understand the intensity levels of brain matter and the tumor substance. Going through the papers of actual researchers from the field to better understand how to operate on the scans ourselves, we also looked for datasets to work on, via which we found a website called OpenNeuro, which helped us get the dataset we have used throughout till now.

For Week 2, We tried to iterate through multiple different scans from the dataset, we tried a few different datasets too, we tried to apply our own inference on those images, trying to use our understanding of which scans actually had a tumor and which ones did not, we were also looking into selecting a base scan for the purpose of comparing it with other scans to better detect a tumor.

Later in the Week, one of our group members found a software MRIcroGL which basically allows people to open MRI Scans directly, it was an alternative for us to see the scans without using a python script, because the scans were a time series of the entire time the person was being scanned, having the third dimension based on that, instead of colour. This also relaxed our work for the week as we could iterate through various slices within the same series to better understand how the scans also progress with time.

For Week 3, We actually started building our coding base, We started working with Python Programming to go through multiple test cases while also learning more from pre existing

literature on the topic, we understood that multiple scans would be required to make a more educated guess on the scans with a tumor, but computation was also slowly becoming more and more costly, as the number of scans increased.

We also started doing calculation based on the regions we found most probable of having a tumor, and according to that we started looking into the voxels based on every scan and accordingly, we realised that our current computation would not allow us to comprehend the entire dataset, we tried to change our approach by making it more GPU oriented, trying to use them instead to cover up for the processing power of each scans, but on further work we realised that even that VRAM was not enough to deal with the huge datasets we were working with, and realised that we might have to try something completely different.

For Week 4, We started looking into ways to convert our image to binary when our scans first started as float64, we converted it to unsigned int16, and now we were converting it to binary, so that we could carry over the data to C++ as we did not have an idea on how to do image processing via that language, so we started looking into two things parallely, how to properly balance out Python and C++ oriented approach or if we could change our language, we are currently aiming for a more balanced approach.

We also tried to apply the forms of filtering that we learnt in class, to see if they actually helped us separate out the tumor in the scans, but none of them ended being well fit for the task, so we realised that the best possible approach was to flip all of this, because number of voxel values was already reaching millions just for one scan, going through each individual reading was bound to be costly, but on practice we noticed that the C++ computation was much more efficient, but sadly we couldn't find ways to involve what we have learnt in the course directly till now during this project.

IV. Literature Review:-

Initially we found a base paper called 'Magnetic resonance image-based brain tumour segmentation methods: A systematic review' via which we better understood how the entire system worked, This itself was a review paper so it helped us look into various methods we could go for, but sadly those methods were more so in the domain of ML & DL, then classical methods.

Yet it turned out to be a great insight for the work we are going to do later in the project, but we also realised the complexity of our topic through this paper, we understood the need to learn some terms associated with the topic, and it was becoming more clearer and clearer that the topic at hand would need quite some reading associated with the terminologies because how the wording was going over our heads when we initially read it.

We then also went through a paper "Rethinking Brain Tumor Segmentation From the Frequency Domain Perspective" which gave us an even greater insight because some more time had passed by and we were getting more comfortable with the language associated with the topic, but the methods they used to achieve this felt way too different to us.

We thought that they would have converted the domain to frequency like we had learnt in class, but it turned out to be much more complex than that. But yet again it helped us gain a different perspective on the same topic, even if the methods felt way more complex than whatever we wanted to do. They were also using Deep Neural Networks, so we at that point realised that we could not try to recreate any of those methods anyways. We are gonna name a few of the methods that they used, Dual-Tree Complex Wavelet Transform (DTCWT) and Nonsubsampled Contourlet Transform (NSCT) via Multi Modality Models, whereas we were still grasping on how to go through a Single Modality Model. They also used the Laplacian Convolution Model which increased our morale to an extent.

The Other Paper we read parallely was more so to acquire another dataset to work on, they had collected many samples of MRI scans too, which we felt would be helpful to run our own code and inference on so that we could better understand our own progress on the same. We also tried to base our workings in the later week on the paper, "Comparative Study of Classical Image Processing Techniques for Brain Tumor Segmentation in MRI Scans," which basically we have tried to recreate to an extent in our works.

V. Results:-

Please find all associated results down below through the work that we have put over multiple weeks of this Project:



Fig. 1. Result of base filtering applied to the MRI scans. The report notes that a tumor is considered an edge-difference anomaly in the brain.

This image contains the RAW binary voxel data of .nii.gz file.

Since we want to ensure that the results are correct, we are performing multiple rounds of baseline and multiple rounds of testing.

For efficient computation, we are implementing C++ alongside Python.

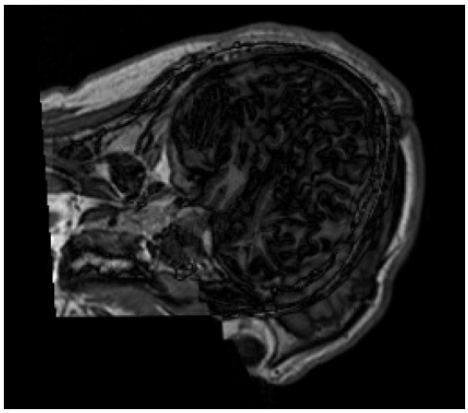


Fig. 2. Attempt to use a healthy MRI scan as a kernel and superimpose it on other scans. The challenge is to map two scans precisely.

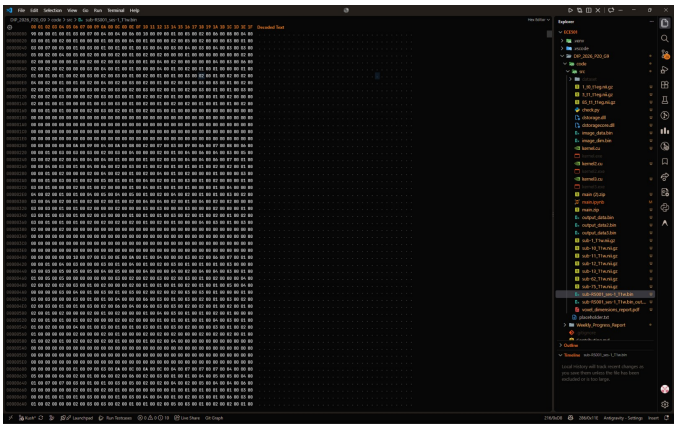


Fig. 3. RAW binary voxel data from a .nii.gz file.

Python helps us open the MRI file and perform basic scaling and alignment.

However, due to the computational limits of python and the limits of python to process multiple rounds fast and efficiently. Our main part of processing is written in C/C++ with the CUDA libraries.

Final image rendering will be done in Python.

VI. What We plan to do Next:-

Currently we are working on converting our python code's data into binary so that it can be processed in C++, once we get those computational results, we are trying to work on whether to continue further on C++ or return the data to Python because the methods that we have learnt are on the basis on Python, so we are reading and looking into it all accordingly, we feel that through this conversion, we are much closer to the segmentation of the tumor, which should be done in two weeks or so, after which we are going to work on image restoration and recovery.

This will allow the patients to better understand the region of tumor growth, and it would also help the doctor to better explain the situation to their patients, they are better professionals on that aspect than us, but if we simplify and shorten the process, it would save time, which is the utmost for the patient at hand.

We also see the need to look into better/alternative methods for what we have done in our project till now, just like how we discovered that the programming language could save us computation, we then plan to make an Interactive UI so that any user could have the simplicity and comfort of going through the scans rather than running the code themselves, creating a more application oriented end for the project, which aims for simplicity and relaxed usage.

VII. Individual Contributions:-

Kush Kelaiya: Majority of the Code Writing, Literature Review, Methodology and some parts of the Documentation.

Rushin Modi: Some of the Code Writing, Literature Review, Discovery of Datasets and Methodology, Some of the Documentation.

Kritee Shah: Some of the Code Writing, Literature Review, Methodology, Literature Review and some of the Documentation.

Jash Odedara: Code Verification and debugging, Literature Review, Methodology, Communication & Conceptualisation and Documentation.

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